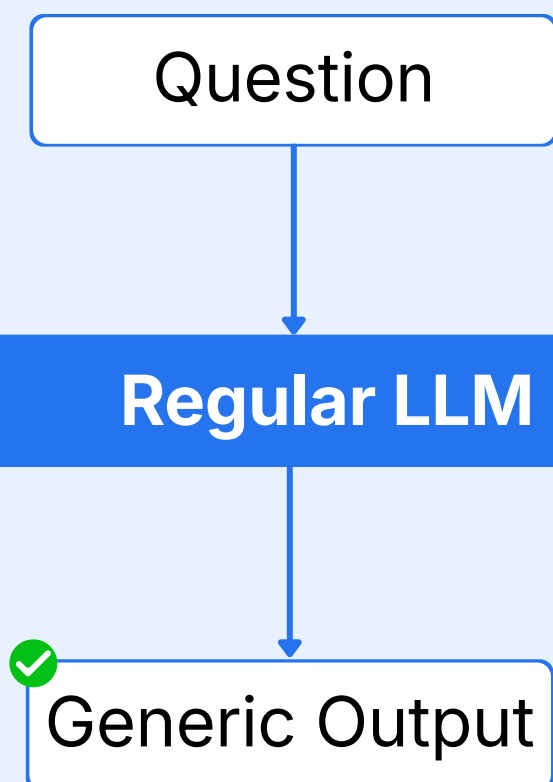
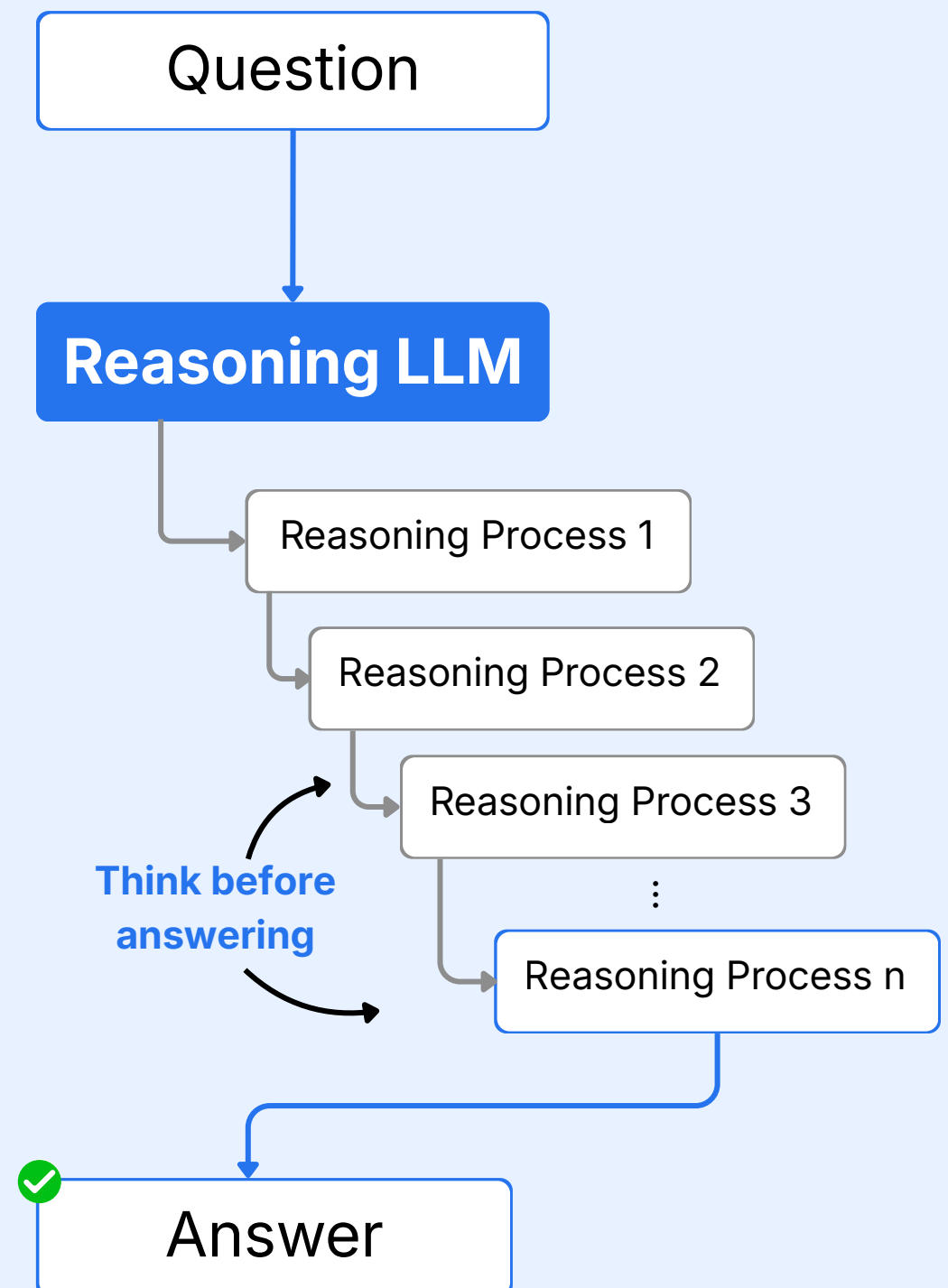


A Guide to Reasoning LLMs vs Regular LLMs

Regular LLM



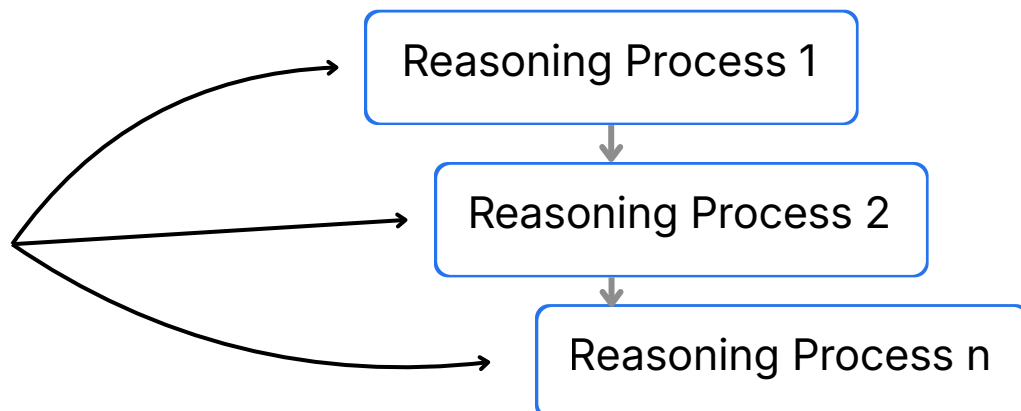
Reasoning LLM



What is Reasoning?

It is a process of answering questions that involve

- complex
- multi-step reasoning



with intermediate stages.

Reasoning question will look something like this

Reasoning Question



A farmer has chickens and cows. There are 30 heads and 100 legs in total. How many chickens and how many cows are there?

These are not reasoning questions

Non - Reasoning Question



What is the capital of France

Non - Reasoning Question



What does CPU stand for?

Response Differences

Reasoning Question

A farmer has chickens and cows. There are 30 heads and 100 legs in total. How many chickens and how many cows are there?

Let's pass this question to both LLM

Regular LLM

Plain Response

Reasoning LLM

Reasoning Process 1

Reasoning Process 2

Reasoning Process 3

⋮

Reasoning Process n

Think before
answering

Answer



A generic LLM can still answer this, but it may respond differently than a reasoning-focused model. In some cases, you can elicit more deliberate reasoning from a generic LLM by **prompting** it to **think step by step**.

Regular LLM style response

Let c be chickens and w be cows.

$c + w = 30$ and $2c + 4w = 100$

Solving gives $w = 20$ and $c = 10$.

So there are 10 chickens and 20 cows.

What's happening here:

- It recognizes a **familiar algebra pattern**
- It compresses the **reasoning into a short answer**
- Intermediate **thinking is implicit** or skipped
- It prioritizes **speed** and **conciseness**

Sometimes a generic model may even guess or shortcut if the pattern is slightly unusual.

Pattern Recognition



Compact Answer

Reasoning LLM style response

A reasoning LLM tends to unpack the logic step by step:

Each animal has one head, so the total number of animals is 30.

If all 30 were chickens, there would be $30 \times 2 = 60$ legs.

But there are actually 100 legs, which is 40 more legs than that.

Each cow contributes 2 extra legs compared to a chicken.

So $40 \div 2 = 20$ cows.

The remaining $30 - 20 = 10$ are chickens.

Therefore, there are 10 chickens and 20 cows.

What's happening here:

- It builds a **chain of logical steps**
- It explains why **each step makes sense**
- It checks **internal consistency**
- The reasoning is **transparent** and **traceable**

Structured
Decomposition



Explicit Logic



The reasoning model is simply optimized to show its work and stay reliable on harder, multi-step problems.

Regular Large Language Model

Regular LLMs are trained with **supervised fine-tuning(SFT)** to learn how to respond to instructions directly.

Examples of Regular LLMs



Reasoning Large Language Model

Reasoning LLMs are trained with supervised fine-tuning and reinforcement learning methods (such as **GRPO**) to respond to instructions by performing step-by-step reasoning before generating an answer.

Examples of Reasoning LLMs

